MOVA One – AI Personal Companion Robot



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Executive Summary

MOVA Robotics S.R.L. is a fictional Paraguayan technology company created to demonstrate how a complete STEM business proposal can be developed and formatted in Microsoft Word. Its first product, MOVA One, is a compact personal AI companion robot designed to combine conversational artificial intelligence, voice interaction, facial recognition, mobility, obstacle detection and a modular magnetic docking system. The robot is designed for students, young professionals, families and technology enthusiasts who want an interactive device that can accompany them on a desk, in a room or through a specially designed magnetic accessory attached to a backpack or other surface.

MOVA One is positioned between a conventional smart assistant and a consumer companion robot. Its main differentiator is the combination of mobility, AI interaction and a modular magnetic attachment system. The product will be offered initially through online sales, technology events, school demonstrations and selected retail partners in Asunción and Gran Asunción. The proposed introductory price is Gs. 1,950,000.

The company will begin with small-batch production and an initial target of 25 units in the first month, increasing monthly sales gradually as awareness grows. The business model includes direct sales, accessories, replacement parts and future software-based services. The initial investment will cover prototyping equipment, electronics, 3D printing, computing equipment, workspace setup, legal formation, initial inventory and working capital.

MOVA Robotics will also apply circular-economy principles by designing the robot to be repairable and modular. The battery, wheels, sensors, external shell and selected electronic modules will be replaceable rather than requiring the disposal of the complete robot. The company will also establish a take-back program for damaged units and reusable electronic components.

The project is fictional and the financial figures are illustrative estimates created for educational purposes. They demonstrate how students could complete the business-plan format; they are not a real commercial forecast.

Resumen Ejecutivo

MOVA Robotics S.R.L. es una empresa tecnológica paraguaya ficticia creada para demostrar cómo se puede desarrollar y presentar un proyecto empresarial STEAM completo. Su primer producto, MOVA One, es un pequeño robot asistente personal que combina inteligencia artificial conversacional, interacción por voz, reconocimiento facial, movilidad, detección de obstáculos y un sistema modular de acoplamiento magnético.

El producto está dirigido principalmente a estudiantes, jóvenes profesionales, familias y personas interesadas en la tecnología que buscan un dispositivo interactivo capaz de acompañarlos sobre un escritorio, dentro de una habitación o mediante un accesorio magnético diseñado para mochila u otras superficies.

MOVA One se posiciona entre un asistente inteligente convencional y un robot de compañía. Su principal diferenciación es integrar movilidad, inteligencia artificial e interacción física en un dispositivo compacto. El precio de venta propuesto para este modelo académico es de Gs. 1.950.000.

La empresa comenzará con producción en pequeños lotes, una estrategia de venta directa y presencia en redes sociales, eventos tecnológicos y demostraciones educativas. El proyecto incorpora economía circular mediante un diseño modular que permite reparar o reemplazar determinadas piezas sin descartar todo el robot.

Company Description Worksheet

Table 1

Company description sheet

Field	MOVA Robotics S.R.L.
Business Name	MOVA Robotics S.R.L.
Company Mission Statement	To create accessible personal robots that combine artificial intelligence, robotics and human-centered design to make technology more useful, interactive and enjoyable.
Company Philosophy / Values	Innovation: We transform advanced technology into useful experiences. Responsibility: We design technology with privacy, safety and repairability in mind. Learning: We promote STEM education and continuous experimentation. Sustainability: We favor modular products that can be repaired, upgraded and reused.
Company Vision	To become a recognized Paraguayan technology company in consumer robotics and AI-assisted products, developing accessible solutions for education, personal assistance and everyday interaction.
Goals & Milestones	Short term: complete the first functional prototype; validate movement and voice interaction; estimate production cost; test the magnetic docking system; conduct market research. Medium term: launch the first commercial batch; establish online sales; build a local support process; release accessories. Long term: expand the product line, improve autonomous navigation, develop educational versions and enter regional markets.
Target Market	Primary: secondary and university students, young professionals and technology enthusiasts in Asunción and Gran Asunción. Secondary: families, schools, technology centers and companies interested in interactive demonstration devices.
Industry / Competitors	The product competes with consumer companion robots and AI desktop robots such as AIBI and EMO from LivingAI and Loona from Loona Inc. MOVA seeks to differentiate itself through local positioning, modular repairability and a magnetic attachment system designed for mobile use.
Legal Structure / Ownership	MOVA Robotics will be presented as a Sociedad de Responsabilidad Limitada (S.R.L.) for this academic model. The five fictional founding partners will contribute capital in equal proportions, subject to the formal requirements that would apply if the company were actually established.

Note. Author's own work.

Products & Services Description

Table 2

Product description sheet

Field	MOVA One
Product / Service Idea	A compact AI personal companion robot that can move, interact by voice, recognize its user, detect obstacles, display expressions and attach to a specially designed magnetic accessory.
Special Benefits	Combines AI conversation with physical mobility; offers an engaging alternative to a static voice assistant; can accompany the user on a desk or through a magnetic accessory; designed for modular repair and upgrades.
Unique Features	Compact wheeled platform; microphone array; camera; speaker; expressive LED/display face; obstacle sensors; Wi-Fi/Bluetooth; rechargeable battery; modular shell; magnetic docking accessory; companion mobile app.
Limits and Liabilities	Battery life is limited; autonomous movement depends on the environment; AI responses may be incorrect; cloud features require connectivity; camera and microphone require privacy controls; the prototype does not replace professional medical, security or safety systems.
Production and Delivery	Electronics are assembled and tested locally; the external shell is produced with 3D printing in the initial stage; firmware is programmed and tested; final units are inspected, packaged and delivered through online orders, events or selected retail partners.
Suppliers	Electronic components, motors, sensors, batteries and connectors will be purchased from specialized electronics suppliers. 3D-printing filament, packaging and mechanical hardware will be purchased locally when available. Specialized components may be imported through distributors.

Note. Author's own work.

Intellectual Property

The MOVA name, logo, visual identity and original product design will be treated as intellectual-property assets of the fictional company. Before real commercialization, the company would verify name availability and consider the appropriate trademark, industrial-design and software protections.

Special Permits

For the academic model, no special product-health registration is assumed. A real commercialization process would need to verify electrical, radio-frequency, consumer-product, battery transportation, data-protection and other applicable requirements before sale.

Product / Service Description

MOVA One is a small interactive robot designed to make AI tangible. Instead of interacting with an assistant only through a phone or smart speaker, the user interacts with a physical object capable of moving, looking, listening, speaking and expressing simple reactions. The product is intended to create a friendly interface between users and AI.

Circular Economy

MOVA One will incorporate circular-economy principles through a modular design. “The external shell, wheels, battery, some sensors and other repairable components will be designed to be replaceable. This approach can extend the product's useful life and reduce electronic waste” (Ellen MacArthur Foundation, 2013).

- Repair instead of replacing the complete robot.
- Offer replacement shells and mechanical modules.
- Recover damaged units for parts.
- Use recyclable cardboard packaging and minimize unnecessary plastic.
- Design accessories so that they remain compatible with future MOVA versions when technically possible.
- Establish a return program for end-of-life units.

Marketing Plan

Market Research

For this sample project, a fictional survey of 100 respondents is used to demonstrate how market research could be presented. The results are illustrative and should not be presented as real research.

Table 3

Market survey results (illustrative data)

Survey question	Illustrative result
Would you be interested in a small AI companion robot?	Yes 72% | Maybe 18% | No 10%
Which feature is most attractive?	Conversation 32% | Movement 25% | Personalization 18% | Reminders 15% | Games 10%
Would you use a robot that can follow you on a desk or room?	Yes 68% | Maybe 22% | No 10%
Would a magnetic attachment system interest you?	Yes 61% | Maybe 24% | No 15%
What price range seems reasonable?	Gs. 1.0–1.5 M 20% | 1.5–2.0 M 46% | 2.0–2.5 M 24% | >2.5 M 10%
Would you buy locally if technical support were available?	Yes 74% | Maybe 18% | No 8%
What concerns you most?	Price 31% | Privacy 26% | Battery 18% | Durability 15% | AI accuracy 10%
Would you recommend it if the product worked as demonstrated?	Yes 70% | Maybe 23% | No 7%
Where would you prefer to buy?	Online 49% | Technology store 28% | Events 15% | School/university 8%
Would you pay for future accessories?	Yes 54% | Maybe 31% | No 15%

Note. Author's own work.

Barriers to Entry

- High initial cost of electronics, batteries, sensors and prototyping equipment.
- Need for specialized knowledge in robotics, embedded programming and AI integration.
- Competition from established international brands.
- Difficulty maintaining consistent quality in small-batch manufacturing.
- Dependence on imported components and exchange-rate variation.
- Customer concerns about privacy, safety and reliability.

Threats and Opportunities

Table 4

Opportunities and threats

Opportunities	Threats
Growing interest in AI companions and consumer robotics.	Rapid technological obsolescence.
STEM education and school robotics programs.	Large international competitors with greater resources.
Local technical support as a differentiator.	Component shortages or price increases.
Accessories and future software services.	Privacy or cybersecurity incidents could damage trust.
Regional expansion in the Southern Cone.	Customers may prefer smartphones or cheaper smart assistants.

Note. Author's own work.

SWOT Analysis Worksheet

Table 5

SWOT Analysis by business area

Business area	Strengths	Weaknesses	Opportunities	Threats
Product / Service Offering	Physical AI interaction; mobile robot; magnetic modular accessory.	Complex electronics; limited battery life in first version.	Interest in AI companions and educational robotics.	Competitors can introduce similar features quickly.
Brand / Marketing	Strong visual identity and demonstrable product.	New brand with no established reputation.	Social media demonstrations are highly visual.	Negative reviews can spread quickly.
Staff / HR	Multidisciplinary team combining programming, electronics, design and business.	Limited experience with mass production.	Training and partnerships with universities or makerspaces.	Skill gaps can delay development.
Finance	High value per unit and potential accessory revenue.	High initial investment.	Preorders and small-batch production can reduce inventory risk.	Exchange-rate and component-price volatility.
Operations / Management	Modular production allows testing by subsystem.	Quality control is technically demanding.	Standardized assembly and test procedures.	Defective electronics may be costly to replace.
Market	Novelty and strong demonstration value.	Premium price limits the market.	AI and robotics are attractive to young consumers.	Low-cost alternatives and imported products.

Note. Author's own work.

Can any of your strengths help with improving your weaknesses or combating your threats?

Yes. The multidisciplinary team can reduce technical weaknesses through structured testing, documentation and continuous learning. Modular design can reduce repair costs and make the company less vulnerable to component failures. Strong demonstrations and transparent communication can help the new brand compete against better-known products.

Based on the information above, what are your immediate goals/next steps?

Complete the functional prototype, validate the movement system, test the magnetic accessory, conduct user testing, estimate the real unit cost and prepare a controlled pilot batch.

Based on the information above, what are your long-term goals/next steps?

Expand the product line, improve autonomous navigation, add educational capabilities, develop accessories and establish regional distribution in Paraguay and neighboring markets. (OpenAI, 2026)

Factors Affecting Product Supply and Demand

Supply will depend on the availability and cost of microcontrollers, motors, sensors, batteries, cameras, speakers, 3D-printing filament, magnets, connectors and packaging. “Production capacity will also depend on assembly time, testing capacity, equipment availability and access to technical personnel. These factors are related to the management of materials, production capacity and operational resources” (Heizer, 2017).

Demand will be influenced by price, perceived usefulness, AI quality, battery life, design, reliability, privacy, availability of technical support, social-media exposure and the novelty of consumer robotics. Demand may increase during school technology fairs, gift seasons and technology events.

Product/Service Features and Benefits

Table 6

Product features and benefits

Feature	Benefit
Voice interaction	Allows natural commands and conversational interaction.
Camera and recognition	Enables personalized interaction and selected visual functions.
Obstacle sensors	Reduces collisions during movement.
Expressive face/display	Makes interaction easier to understand and more engaging.
Magnetic docking accessory	Allows the robot to travel with the user on compatible surfaces.
Modular components	Supports repair and upgrades.
Mobile application	Provides configuration, updates and additional controls.
Rechargeable battery	Allows portable use without permanent cable connection.

Note. Author's own work.

Target Customer

The primary target customer is a technology-oriented student or young adult between approximately 15 and 30 years old living in Asunción or Gran Asunción, with interest in AI, gadgets, gaming, robotics or personalized technology. Secondary customers include families, educators, technology centers and companies seeking interactive demonstration devices.

“The target market is also supported by the growing interest in AI-powered consumer experiences in Latin America. According to IBM, 29% of surveyed consumers in the region had already used AI for shopping, while many others expressed interest in using AI for product research, customer service, purchasing and personalized recommendations. This trend supports the potential appeal of a personal robotic device that combines artificial intelligence, personalization and interactive technology” (IBM, 2024).

For Consumers

Consumers receive an interactive personal device that combines entertainment, experimentation and practical AI functions. The product can be used as a desk companion, conversational assistant, reminder device and robotics platform.

“Consumers receive an interactive personal device that combines entertainment, experimentation and practical AI functions. The product can be used as a desk companion, conversational assistant, reminder device and robotics platform” (OpenAI, 2026).

For Businesses

MOVA can offer customized units to technology companies, educational institutions, event organizers and businesses that want an interactive technological demonstration or promotional device. Institutional versions may include branding, preconfigured behaviors and educational software.

Key Competitors

The competitive set for this academic project includes companion robots and desktop robots with artificial intelligence already available on the market. The products AIBI, EMO and Loona are used as real references. The following comparison is qualitative; commercial prices and specifications should be verified before any real business decision.

Table 7

Comparison with reference competitors

Competitor	Reference characteristics	MOVA differentiation
AIBI – LivingAI	Portable AI companion; face recognition; voice commands; ChatGPT-connected Q&A; reminders and interactive behaviors.	MOVA emphasizes physical mobility, modular repair and a dedicated magnetic transport accessory.
EMO – LivingAI	AI desktop pet with movement, sensors, recognition, expressions and smart-home-related functions.	MOVA is designed from the beginning around portability and modular attachment rather than only desktop companionship.
Loona – Loona Inc.	Consumer AI companion robot with expressive interaction, navigation and companion-oriented features.	MOVA focuses on a smaller initial scope, local support and a modular ecosystem aimed at the Paraguayan market.

Note. Competitor characteristics adapted from information published by **(LivingAI, 2026)** *and* **(Loona Inc., 2026)**. *The "MOVA differentiation" column corresponds to the project's analysis and proposal.*

Competitor Data Collection Plan

Table 8

Comparative analysis of competitive factors

Factor	AIBI	EMO	Loona
Price	Premium consumer robot; verify current official price.	Premium consumer robot; official store currently lists EMO at US$279.	Premium consumer robot; verify current official price.
Benefits / Features	Voice interaction, face recognition, reminders and AI Q&A.	Movement, sensors, recognition, expressions, AI interaction and desktop navigation.	Companion interaction, movement/navigation and evolving AI features.
Size / Profitability	Portable form factor; premium positioning.	Desktop-sized; premium positioning.	Larger companion robot; premium positioning.
Market Strategy	Online direct-to-consumer and technology-oriented community.	Online direct-to-consumer and social/community positioning.	Online direct-to-consumer and global companion-robot positioning.

Note. Adapted from public information from (LivingAI, 2026) and (Loona Inc., 2026).

Competitive Analysis

Table 9

Strengths and weaknesses — advertising

FACTOR	Importance to Customer 1–5	MOVA Robotics S.R.L.	AIBI Pocket	Eilik	Loona
Products	5	S	S	S	S
Price	4	S	S	S	W
Quality	5	S	S	S	S
Service	3	W	S	S	S
Reliability	5	W	S	S	S
Stability	4	W	S	S	S
Expertise	4	W	S	S	S
Company Reputation	3	W	S	S	S
Location	2	S	W	W	W
Sales Method	4	W	S	S	S
Advertising	3	W	S	S	S

Note. Adapted from public information from (LivingAI, 2026) and (Loona Inc., 2026).

Competitive Analysis Worksheet

Table 10

Competitive analysis summary matrix

Factor	Importance to Customer (1–5)	MOVA	AIBI	EMO	Loona
Products	5	S	S	S	S
Price	5	S	W	W	W
Quality	5	W	S	S	S
Service	4	S	W	W	W
Reliability	5	W	S	S	S
Stability	4	W	S	S	S
Expertise	4	W	S	S	S
Company Reputation	3	W	S	S	S
Location	3	S	W	W	W
Sales Method	3	S	S	S	S
Advertising	3	W	S	S	S

Note. Adapted from public information from (LivingAI, 2026) and (Loona Inc., 2026).

Positioning / Niche

MOVA will occupy a niche between static smart assistants and high-priced imported companion robots. The initial niche is technology-oriented young consumers who want a physical AI experience and value local support. The brand will be positioned as friendly, modern, experimental and repairable.

The product's strongest visual and functional differentiator is its modular magnetic docking system. Instead of remaining only on a desk, MOVA can be attached to compatible accessories and transported with the user. The system will be designed to be secure during normal use while explicitly excluding climbing, dangerous surfaces or situations where a falling robot could cause injury. (OpenAI, 2026)

How You Will Market Your Product/Service

MOVA will use a mix of short-form video, demonstrations, website/e-commerce, technology events, school demonstrations, creator collaborations and product trials. The strategy will focus on showing the robot performing real tasks instead of only using static advertising.

Promotional Mix / Communication Plan – 50 Weeks

Table 11

Communication and promotion plan — 50 weeks

Media / Tool	Channel / Means	Purpose	Type	W4	W5	W6	W7	W8	W9	W10	W11	W12
Instagram Reels	Digital / direct	Short demonstrations	Ongoing	X		X		X		X		X
TikTok videos	Digital / direct	AI interaction and movement	Ongoing	X	X	X	X	X	X	X	X	X
Instagram Feed/Stories	Digital / direct	Product updates and education	Ongoing	X	X	X	X	X	X	X	X	X
Company Website	Digital / direct	Product, support and orders	Before startup + ongoing	X	X	X	X	X	X	X	X	X
Online Store	Digital / direct	Sales and customer service	Ongoing	X	X	X	X	X	X	X	X	X
Landing Page	Digital / direct	Campaign-specific information	Campaign	X	X	X	X	X				
Printed Flyers	Digital / direct	Events and technology fairs	Ongoing	X				X				X
Business Cards / QR	Digital / direct	Direct networking	Before startup + events	X								X
Branded Stickers	Digital / direct	Packaging and events	Ongoing	X				X				X
Product Demonstrations	Digital / direct	Live robot demonstrations	Ongoing			X				X		
Giveaways	Digital / direct	Audience growth	Ongoing									X

Note. Author's own work.

What image do you want to project for your business brand?

MOVA wants to project an image that is intelligent, friendly, modern, curious and trustworthy. The brand should make advanced robotics feel approachable rather than intimidating.

What design elements will you use to market your business? Explain how they support your brand

The brand identity will use a compact geometric logo based on the letters MOVA and a simplified robot-eye symbol. The visual system will combine dark graphite, electric blue, white and a bright accent tone. The product photography will use clean backgrounds and close-up shots of the robot's face, sensors and magnetic accessory. The same identity will be used on packaging, website, social media, event banners and app screens.

The proposed slogan is “Small robot. Big intelligence.” It communicates the main positioning: a physically small device that offers a richer interaction than a conventional static assistant.

Promotional Budget

The following budget is an illustrative estimate for the academic model. Before-startup promotion is estimated at Gs. 3,250,000. Ongoing promotion is estimated at Gs. 1,450,000 per month, or Gs. 17,400,000 per year. The estimated first-year promotional budget is therefore Gs. 20,650,000.

Table 12

Marketing expenses budget

Media / Tool	Channel / Means	Budget Type	Cost per Unit (Gs.)	Quantity / Month	Estimated Cost (Gs.)	Evidence / Basis
Social Media Marketing	TikTok videos	Ongoing	0	8	0	Created internally
Social Media Marketing	Instagram Reels	Ongoing	0	8	0	Created internally
Social Media Marketing	Paid social ads	Ongoing	350000	2	700000	Illustrative monthly estimate
Website / E-commerce	Website	Before startup	0	1	0	Created with internal tools
Website / E-commerce	Landing page	Before startup	0	1	0	Created internally
Print Advertising	Flyers	Ongoing	12000	50	600000	Illustrative printing estimate
Print Advertising	QR cards	Before startup	1000	300	300000	Illustrative printing estimate
Product Sampling	School/event demos	Ongoing	150000	2	300000	Transport and demo materials
Content Marketing	Creator collaborations	Before startup	650000	3	1950000	Three prototype/demo units
Content Marketing	Giveaways	Ongoing	1050000	1	1050000	One demo unit every selected campaign

Note. Author's own work.

Marketing Expenses Strategy Chart

The chart below uses the same activities as the communication plan and separates one-time and ongoing spending.

Table 13

Summary of marketing activities and monthly costs

Activity	Action / Description	Frequency	Estimated monthly cost
Paid social advertising	Promote demonstration videos and product launches.	2 campaigns/month	Gs. 700,000
Flyers	Technology fairs and school demonstrations.	50 units/month	Gs. 600,000
QR cards	Direct customers to website and support.	Startup batch	Gs. 300,000 startup
Creator collaborations	Provide demonstration units for review.	Campaign	Gs. 1,950,000 startup
Demonstration events	Transport and materials for live demonstrations.	2/month	Gs. 300,000
Giveaways	Audience-growth campaigns.	1/month selected periods	Gs. 1,050,000

Note. Author's own work.

Pricing Strategies

Name of the company: MOVA Robotics S.R.L.

MOVA will use a cost-plus and value-based pricing approach. The illustrative production cost of one MOVA One is Gs. 1,050,000. A proposed selling price of Gs. 1,950,000 provides a contribution of Gs. 900,000 per unit before fixed operating costs. The price reflects the electronics, mechanical assembly, AI integration, testing, warranty provision and local support included in the product.

The company will not compete only by being the cheapest option. Instead, it will compete through local support, repairability, modular design, the magnetic accessory and a clear user experience.

Distribution Channel Assessment Worksheet

Table 14

Distribution channel assessment

Factor	Online direct sales	Technology stores	Events / school demonstrations
Ease of Entry	High: website, social media, payment and delivery process.	Medium: requires retailer agreement and margin.	High: requires event permission and transport.
Geographic Proximity	Can reach customers throughout Paraguay.	Good in Asunción and major cities.	Excellent for direct demonstrations in target communities.
Costs	Low-medium: platform, payment and delivery fees.	Higher: retailer margin, stock and logistics.	Medium: transport, stand and demo costs.
Competitors' Positions	Many imported products already sell online.	Established brands have retail relationships.	Few customers can test premium robots locally.
Management Experience	Requires order, payment and support management.	Requires negotiation, stock and invoices.	Requires event organization and demonstrations.
Staffing Capabilities	One person can manage online orders initially.	Requires commercial relationship management.	Two or three people can demonstrate and sell.
Marketing Needs	Video demonstrations, reviews and social proof.	Point-of-sale material and product training.	Live demonstrations, flyers, QR codes and social posts.

Note. Author's own work.

Pronóstico de Ventas de 12 Meses / 12-Month Sales Forecast

Para la proyección académica, la empresa comienza con 25 unidades en enero y aumenta las ventas en cinco unidades por mes a medida que mejora el conocimiento de la marca y la distribución. El precio de venta propuesto se mantiene en Gs. 1.950.000 por MOVA One.

Table 15

Pronóstico de ventas mensual — 12 meses

Mes	Unidades vendidas	Precio / unidad (Gs.)	Ventas (Gs.)
Enero	25	1950000	48750000
Febrero	30	1950000	58500000
Marzo	35	1950000	68250000
Abril	40	1950000	78000000
Mayo	45	1950000	87750000
Junio	50	1950000	97500000
Julio	55	1950000	107250000
Agosto	60	1950000	117000000
Septiembre	65	1950000	126750000
Octubre	70	1950000	136500000
Noviembre	75	1950000	146250000
Diciembre	80	1950000	156000000

Nota. Elaboración propia.

Unidades proyectadas anuales: 630. Ventas proyectadas anuales: Gs. 1.228.500.000.

Presupuesto de Ventas

Table 16

Resumen del pronóstico anual de ventas

Producto / Servicio	Referencia mensual	Proyección anual
MOVA One	25 unidades en el mes de lanzamiento	630 unidades
Precio por unidad	Gs. 1.950.000	Gs. 1.950.000
Ventas	Gs. 48.750.000 en enero	Gs. 1.228.500.000

Nota. Elaboración propia.

Plan Operativo

Producción

El proceso de producción comenzará con la inspección de recepción de los componentes electrónicos y mecánicos. El equipo ensamblará el chasis, instalará los motores y las ruedas, conectará los sensores, instalará la cámara, el micrófono y el altavoz, ensamblará el sistema de batería, montará la carcasa externa, cargará el firmware y realizará pruebas funcionales.

1. Recibir y clasificar los componentes electrónicos, mecánicos y de embalaje.
2. Inspeccionar los componentes para detectar daños visibles y verificar las cantidades.
3. Ensamblar el chasis mecánico, los motores y las ruedas.
4. Instalar el controlador, los sensores, la cámara, el micrófono y el altavoz.
5. Instalar y asegurar la batería recargable y los componentes de protección.
6. Instalar la carcasa externa y la interfaz de acoplamiento magnético.
7. Cargar el firmware y configurar el dispositivo.
8. Vincular el robot con la aplicación móvil.
9. Realizar pruebas de movimiento, audio, cámara, sensores y batería.
10. Realizar una inspección final de calidad y registrar el número de serie.
11. Empaquetar el producto con cargador, instrucciones, información de seguridad y condiciones de garantía.
12. Liberar la unidad para la venta.

Control de Calidad

- Verificar la alineación de las ruedas y el movimiento en todas las direcciones programadas.
- Verificar la detección de obstáculos y el comportamiento de parada de emergencia.
- Probar la calidad del micrófono y el altavoz.
- Probar las funciones de cámara y reconocimiento en condiciones controladas.
- Comprobar la carga, la temperatura de la batería y la retención de carga.
- Probar el acoplamiento magnético bajo las condiciones de carga y uso previstas.
- Verificar la versión del firmware y la conectividad.
- Inspeccionar la carcasa externa para detectar grietas, bordes afilados o componentes sueltos.
- Registrar el resultado de cada inspección final por número de serie.

Ubicación

La empresa ficticia operará desde un pequeño taller tecnológico en Asunción, Paraguay, con áreas para ensamblaje electrónico, impresión 3D, pruebas, almacenamiento y administración. Las ventas a clientes serán inicialmente en línea y mediante eventos; no será necesario un local comercial físico al inicio.

Entorno Jurídico

Para este modelo educativo, MOVA Robotics está estructurada como una Sociedad de Responsabilidad Limitada (S.R.L.) con cinco socios fundadores. Los siguientes pasos reproducen el tipo de sección de constitución legal mostrado en el proyecto de referencia. Se presentan como un plan académico y deben verificarse con profesionales paraguayos antes de constituir una empresa real.

Table 17

Procedimientos de constitución legal (modelo académico)

Procedimiento	Plan académico
Constitución por escritura pública – Escribanía	Preparar la escritura de constitución, nombre de la empresa, domicilio legal, objeto social, capital, socios y administración.
Integración del capital social – Entidad bancaria	Aportar el capital acordado en efectivo y/o activos elegibles, documentando cada aporte.
Inscripción en los Registros Públicos	Presentar la documentación requerida para la inscripción de la empresa y su reconocimiento legal.
Publicación de la constitución	Completar la publicación u otro aviso formal requerido por el procedimiento paraguayo aplicable.
Inscripción en la DNIT y obtención del RUC	Registrar la empresa a efectos tributarios y obtener el RUC antes de comenzar las operaciones comerciales formales.
Matrícula de comerciante	Completar los requisitos de inscripción comercial aplicables.
Habilitación de los libros contables	Mantener los registros contables y societarios requeridos por el régimen aplicable.
Inscripción municipal y patente comercial	Completar la inscripción municipal y la autorización comercial correspondiente al lugar de operación.
Inscripción patronal en el IPS	Si se contrata personal, completar la inscripción patronal y las obligaciones de seguridad social aplicables.
Inscripción patronal en el MTESS	Completar las inscripciones patronales y laborales correspondientes.
Registro de la marca en la DINAPI	Evaluar y presentar la solicitud de protección de la marca MOVA y del logotipo.
Permisos adicionales	Verificar los requisitos eléctricos, de radiofrecuencia, baterías, productos de consumo, ambientales y de protección de datos aplicables al producto real.

Nota. Elaboración propia.

Obligaciones Fiscales

La empresa deberá cumplir con el régimen tributario aplicable a su estructura jurídica y actividad reales, incluyendo facturación, contabilidad, declaraciones fiscales e impuestos aplicables. El tratamiento tributario exacto, las tasas y las obligaciones deberán verificarse con la DNIT y un contador paraguayo antes de iniciar operaciones reales.

Personal

Table 18

Hoja de roles y responsabilidades del equipo directivo

Cargo	Responsabilidades	Experiencia	Habilidades	Personalidad	Requisitos físicos	Remuneración ilustrativa
Gerente de Administración	Finanzas, documentos, pagos, coordinación contable y administración diaria.	Se prefiere experiencia en administración de empresas o contabilidad.	Organización, contabilidad, comunicación.	Responsable, organizado, analítico.	Trabajo con computadora y gestión de documentos.	Gs. 5.500.000/mes
Gerente de Ingeniería y Producción	Desarrollo del producto, ensamblaje, pruebas, inventario y control de calidad.	Se prefiere experiencia en electrónica, robótica o producción.	Electrónica, programación, resolución de problemas, trabajo en equipo.	Metódico, curioso, paciente.	Trabajo en taller y uso de computadora.	Gs. 7.000.000/mes
Gerente de Software e IA	Firmware, integración de aplicaciones, servicios de IA, actualizaciones y coordinación de ciberseguridad.	Se prefiere experiencia en programación e IA.	Programación, APIs, depuración, documentación.	Analítico, creativo, persistente.	Trabajo con computadora.	Gs. 7.000.000/mes
Gerente Comercial	Ventas, clientes, promociones, alianzas y estrategia comercial.	Se prefiere experiencia en ventas o marketing.	Comunicación, negociación, atención al cliente.	Amable, proactivo, persuasivo.	Uso de computadora y reuniones con clientes.	Gs. 5.000.000/mes

Coordinador de Diseño y Soporte	Diseño industrial, embalaje, documentación para usuarios y soporte posventa.	Se prefiere experiencia en diseño, comunicación o soporte técnico.	Diseño, documentación, comunicación.	Empático, orientado a los detalles, creativo.	Uso de computadora y trabajo ocasional en taller.	Gs. 4.500.000/mes

Nota. Elaboración propia.

Inventario

MOVA mantendrá registros separados para componentes en bruto, unidades en proceso, robots terminados, repuestos y embalajes. La política de inventario inicial está diseñada para evitar que quede demasiado capital inmovilizado en componentes, manteniendo al mismo tiempo inventario suficiente para respaldar aproximadamente un mes de producción planificada.

Table 19

Inventario inicial proyectado

Categoría de inventario	Objetivo de stock inicial	Almacenamiento / control
Microcontroladores y placas electrónicas	70 unidades	Contenedores seguros para ESD y recipientes etiquetados.
Motores y ruedas	80 juegos	Almacenamiento seco y etiquetado.
Sensores y cámaras	80 juegos	Embalaje protector antiestático.
Baterías	50 unidades	Almacenamiento controlado según las instrucciones de seguridad del proveedor.
Filamento para impresión 3D	25 kg	Almacenamiento seco y protegido del calor excesivo.
Imanes / componentes de acoplamiento	100 juegos	Recipientes etiquetados para piezas mecánicas.
Embalaje	70 kits completos	Estanterías limpias y secas.
MOVA One terminado	10 unidades como mínimo	Área protegida para productos terminados.

Nota. Elaboración propia.

Proveedores

Table 20

Hoja de proveedores

Categoría de proveedor	Materiales / servicios	Criterios de selección	Política de pago
Distribuidor de electrónica	Microcontroladores, sensores, conectores, placas.	Disponibilidad, garantía, soporte técnico, precio.	60 % en efectivo / 40 % dentro de 30 días cuando se negocie.
Proveedor mecánico	Ruedas, engranajes, tornillos, rodamientos, imanes.	Tolerancia, durabilidad, precio, disponibilidad.	Efectivo inicialmente; negociar crédito después de contar con historial de compras.
Proveedor de baterías	Paquetes de baterías recargables y sistemas de protección.	Documentación de seguridad, calidad, garantía.	Efectivo o crédito a corto plazo.
Proveedor de impresión 3D	Filamento y consumibles de impresión.	Consistencia y disponibilidad de materiales.	Efectivo / 30 días.
Proveedor de embalaje	Cajas de cartón, insertos y etiquetas.	Reciclabilidad, calidad de impresión, precio.	30 días cuando sea posible.

Nota. Elaboración propia.

Plan de Compras y Política de Crédito

Para el modelo académico de flujo de efectivo, la empresa supone que el 60 % de las compras a proveedores se paga en el mes de la compra y el 40 % se paga al mes siguiente. Esta es una suposición de planificación y no un acuerdo garantizado con los proveedores.

Política de Crédito

La mayoría de las ventas de MOVA One se pagará al momento de la compra mediante transferencia, tarjeta, billetera digital u otro medio de pago disponible. Los pedidos institucionales o empresariales podrán recibir un cronograma de pagos negociado después de una revisión crediticia básica. Los pedidos institucionales personalizados requerirán un anticipo del 50 % y el saldo restante antes de la entrega.

Management Worksheet

Ana Benítez – Co-founder and Administration Manager

Fictional student profile created for the sample. Interested in entrepreneurship, finance and organization. Responsible for budgeting, documentation and administrative planning.

Gaps in management or experience

Limited experience with real accounting systems and company formation.

Skills

Organization, budgeting, communication, spreadsheet management.

Personality traits

Responsible, analytical, organized.

Advisors

STEM teaching team and external business advisor.

Diego Ferreira – Co-founder and Engineering & Production Manager

Fictional student profile. Interested in robotics, electronics and prototyping. Leads mechanical assembly, electronics integration and quality tests.

Gaps in management or experience

Limited experience with industrial manufacturing and mass production.

Skills

Electronics, robotics, prototyping, troubleshooting.

Personality traits

Curious, methodical, persistent.

Advisors

Robotics/electronics mentor.

Sofía Rojas – Co-founder and Software & AI Manager

Fictional student profile. Interested in programming, AI and app development. Leads firmware, software integration and AI interaction design.

Gaps in management or experience

Limited experience with production-grade cybersecurity and scalable cloud systems.

Skills

Programming, APIs, debugging, AI integration.

Personality traits

Analytical, creative, persistent.

Advisors

Software/AI mentor.

Mateo López – Co-founder and Commercial Manager

Fictional student profile. Interested in communication, technology products and entrepreneurship. Leads sales, customer research and partnerships.

Gaps in management or experience

Limited experience negotiating with professional retailers.

Skills

Communication, negotiation, customer service, sales.

Personality traits

Proactive, confident, collaborative.

Advisors

Marketing/business advisor.

Valentina Giménez – Co-founder and Design & Support Coordinator

Fictional student profile. Interested in design, user experience and technology communication. Leads visual identity, packaging, documentation and support.

Gaps in management or experience

Limited experience with industrial design certification and technical support at scale.

Skills

Design, documentation, UX, communication.

Personality traits

Creative, empathetic, detail-oriented.

Advisors

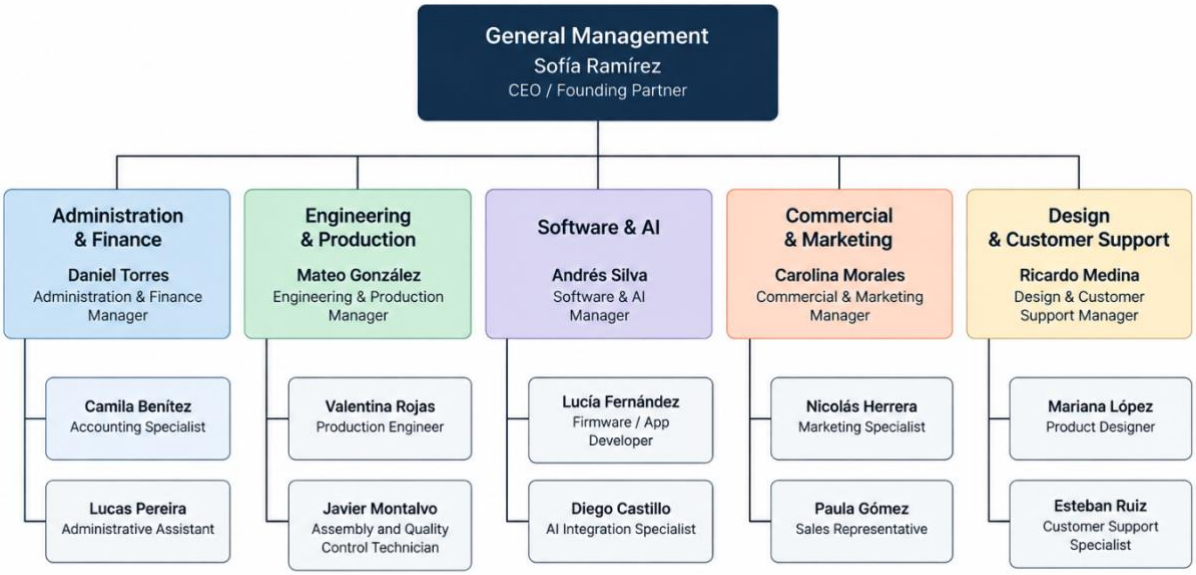
Design/product mentor.

Organization Chart

MOVA Robotics S.R.L. The organizational structure proposed for MOVA Robotics S.R.L. is presented in Figure 1.

Figure 1

Company organizational chart



Gastos Iniciales y Capitalización

Gastos de Puesta en Marcha

Table 21

Gastos de puesta en marcha

Gasto	Costo estimado	Frecuencia
Garantía / depósito del espacio de trabajo	Gs. 4.000.000	Una vez
Primer mes del espacio de trabajo	Gs. 4.000.000	Mensual
Conexión / instalación de servicios	Gs. 1.000.000	Una vez
Servicios legales y de constitución	Gs. 3.500.000	Una vez
Reserva para registro de marca / identidad	Gs. 1.500.000	Una vez
Reserva para seguros	Gs. 1.200.000	Estimación anual
Instalación de internet y comunicaciones	Gs. 400.000	Una vez
Embalaje y etiquetas iniciales	Gs. 1.200.000	Puesta en marcha
Campaña de marketing inicial	Gs. 3.250.000	Puesta en marcha

Nota. Elaboración propia.

Plan de Gastos Generales de la Empresa

Table 22

Plan de gastos generales mensuales

Gasto	Estimación mensual (Gs.)	Frecuencia
Alquiler	4000000	Mensual
Servicios públicos	1200000	Mensual
Internet / comunicaciones	300000	Mensual
Limpieza y suministros del taller	300000	Mensual
Software / nube / servicios de IA	1200000	Mensual
Mantenimiento	500000	Mensual
Seguros	100000	Equivalente mensual
Publicidad	1450000	Mensual
Contabilidad / servicios profesionales	500000	Mensual

Nota. Elaboración propia.

Plan Financiero

Proyección de Pérdidas y Ganancias a 12 Meses

Table 23

Proyección de pérdidas y ganancias — 12 meses

Concepto	Proyección de 12 meses (Gs.)
Ventas	1.228.500.000
Costo de ventas	661.500.000
Ganancia bruta	567.000.000
Gastos operativos	462.600.000
Ganancia antes de impuestos	104.400.000
Reserva ilustrativa de impuesto a la renta (10 %)	10.440.000
Ganancia neta	93.960.000

Nota. Elaboración propia.

La línea de impuestos es una reserva de planificación ilustrativa utilizada para demostrar la estructura del modelo. El tratamiento tributario real debe determinarse de acuerdo con el régimen tributario paraguayo aplicable y las circunstancias reales de la empresa.

Proyección de Flujo de Efectivo

Para simplificar, el modelo académico supone que las ventas a consumidores se cobran en el mes de la venta y que el 40 % de las compras a proveedores se paga al mes siguiente. La tabla es un modelo simplificado del flujo de efectivo operativo.

Table 24

Proyección de flujo de efectivo mensual

Mes	Efectivo inicial	Cobros en efectivo	Salidas de efectivo	Efectivo final
Ene	0	48.750.000	64.800.000	-16.050.000
Feb	-16.050.000	58.500.000	70.050.000	-27.600.000
Mar	-27.600.000	68.250.000	75.300.000	-34.650.000
Abr	-34.650.000	78.000.000	80.550.000	-37.200.000
May	-37.200.000	87.750.000	85.800.000	-35.250.000
Jun	-35.250.000	97.500.000	91.050.000	-28.800.000
Jul	-28.800.000	107.250.000	96.300.000	-17.850.000
Ago	-17.850.000	117.000.000	101.550.000	-2.400.000
Sep	-2.400.000	126.750.000	106.800.000	17.550.000
Oct	17.550.000	136.500.000	112.050.000	42.000.000
Nov	42.000.000	146.250.000	117.300.000	70.950.000
Dic	70.950.000	156.000.000	122.550.000	104.400.000

Nota. Elaboración propia.

Balance Proyectado

Table 25

Balance proyectado

ACTIVO	Gs.
Disponibilidades	104,400,000
Inventario	10,500,000
Activo fijo	59,000,000
Cargos diferidos / preoperativos	13,650,000
TOTAL ACTIVO	187,550,000
PASIVO Y PATRIMONIO NETO	Gs.
Proveedores	22,050,000
Capital aportado	120,000,000
Resultado acumulado / ajuste	45,500,000
TOTAL PASIVO Y PATRIMONIO	187,550,000

Nota. Elaboración propia.

Cálculo del Punto de Equilibrio

Table 26

Cálculo del punto de equilibrio

Concepto	Valor
Costos fijos mensuales	Gs. 38.550.000
Costo unitario variable	Gs. 1.050.000
Precio por unidad	Gs. 1.950.000
Aporte por unidad	Gs. 900.000
Punto de equilibrio	43 unidades de MOVA One/mes
Valor de ventas del punto de equilibrio	Gs. 83.850.000

Nota. Elaboración propia.

La empresa necesita vender aproximadamente 43 unidades de MOVA One al mes para cubrir los costos fijos y los costos unitarios variables modelados. Cada unidad adicional aporta aproximadamente Gs. 900.000 a la ganancia antes de impuestos y otros ajustes no incluidos en el modelo.

Uso del Capital

Table 27

Uso del capital inicial

Categoría	Uso	Inversión (Gs.)
Activos fijos	Impresoras 3D y equipos de fabricación	12000000
Activos fijos	Equipos de prueba electrónica y herramientas	10000000
Activos fijos	Computadoras y redes	15000000
Activos fijos	Instalación y mobiliario del taller	22000000
Preoperativo	Servicios legales / constitución / configuración de marca	8000000
Preoperativo	Desarrollo del prototipo	10000000
Capital de trabajo	Inventario inicial de componentes	25000000
Capital de trabajo	Reserva de efectivo operativo	18000000

Nota. Elaboración propia.

Inversión inicial ilustrativa total: Gs. 120.000.000.

Se supone que la inversión se financia mediante aportes iguales de los cinco socios fundadores ficticios, sin un préstamo bancario en el escenario académico base.

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Appendix A

Prototype Photographs

Figure 2

Front view



Figure 3

Side view



Figure 4

Rear view



Appendix B

Survey Instrument

Figure 5

Question 1

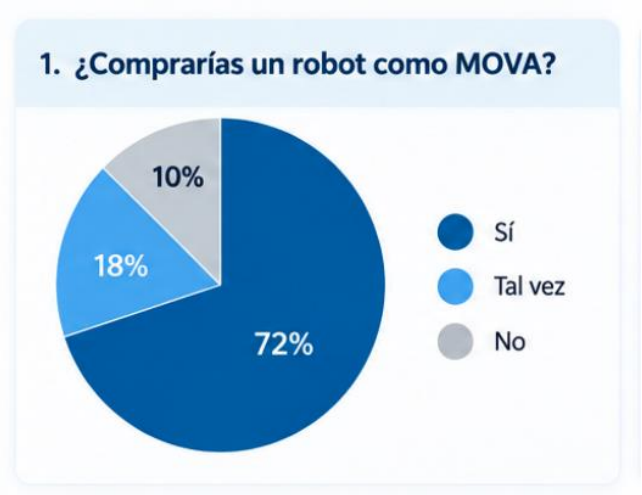
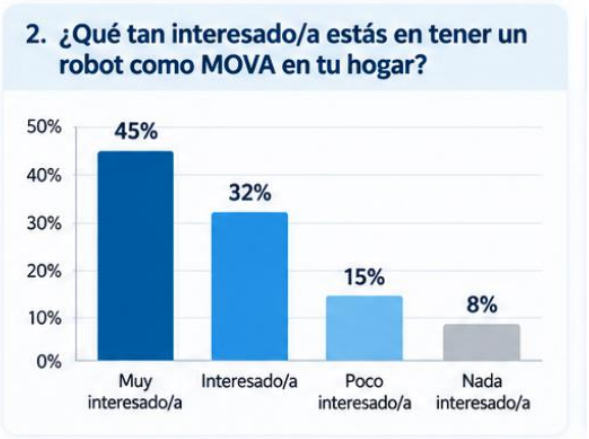


Figure 6

Question 2



Appendix C

Technical Architecture

Table 28

Technical architecture of the product

Subsystem	Example component / function
Controller	Microcontroller or embedded computer
AI connectivity	Cloud AI API and/or local command set
Vision	Camera module
Audio	Microphone array + speaker
Movement	Two or more motors with wheel encoders
Navigation	Distance sensors / obstacle detection
Power	Rechargeable battery + protection circuit
Connectivity	Wi-Fi / Bluetooth
Physical attachment	Magnetic docking system
Software	Firmware + companion application

Note. Author's own work.